Bernice Animwaa Boateng

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Summary

AI/ML engineer and software developer with a degree in computer engineering and hands-on machine learning experience in healthcare. At Really Great Tech I built ML pipelines for clinical signal data, including PAD severity classification, and worked on an AI system that detects distress in phone calls. On my own, I build and deploy full-stack ML apps, from training the model to shipping a live API.

Experience

AI/ML engineer and Data Analyst (National Service) Dec. 2025 – Sep. 2026
Really Great Tech *Accra, Ghana*

Semler Scientific, PAD severity: Built and benchmarked XGBoost and PyTorch models that classify peripheral artery disease severity from arterial PPG waveforms, validated with 6-fold Leave-One-Clinic-Out across 6 clinics (346+ limbs) so each clinic is tested as unseen data.

Engineered and evaluated 111 pulse-waveform biomarkers (pyPPG), used Kernel PCA and metaheuristic feature selection (zoofs) to find compact feature sets, and calibrated thresholds on training folds only, reaching about 80% sensitivity and 79% specificity (ROC AUC 0.82) on unseen clinics.

Built interactive dashboards showing performance sweeps, confusion matrices, 95% confidence intervals and per-clinic results. Earlier on Semler, worked on XGBoost regression predicting myocardial strain from PPG, with feature extraction, recursive feature elimination and leakage-safe hyperparameter tuning scripts.

AppWork VoiceAI: Helped build an AI system for a property management company that flags distress in inbound calls. Benchmarked Hugging Face speech emotion models, labelled distress ground truth from a 911 calls dataset, and worked on a pipeline that fuses audio and text signals.

Software Engineer and IT Intern Apr. 2025 – Aug. 2025
Cocoa Clinic *Accra, Ghana*

Helped maintain and update the hospital's electronic health record (EHR) systems, worked with the IT team on digital transformation projects and internal reporting tools, and handled technical support for software and hardware issues.

Projects

Vellichor AI: Emotion & Theme Analysis | *FastAPI, GPT-2, Hugging Face, Docker* Live App

Fine-tuned GPT-2 end to end to classify the emotion and subtext of written dialogue into 7 categories, served through a FastAPI backend. Used context-aware features and dropout to read subtext instead of surface keywords, and it outperformed standard models on nuanced fiction.

MedMate DDI: Drug Interaction Checker | *Python, React, Django, Transformers, XGBoost* Live App

Co-developed a full-stack app for a three-person final year project that detects drug-drug interactions, using DrugBank and PubChem data.

Built the severity classification models (XGBoost and Random Forest) and led the transformer model that generates plain-language interaction descriptions.

Deployed the Django REST API on Render and the React frontend on Vercel.

CardioGlobal: Heart Disease Data Visualization | *Python, Dash, Plotly, scikit-learn, LightGBM* Live App

Built an interactive dashboard for clinicians and researchers that combines WHO/GHO global mortality data with the UCI Cleveland Heart Disease dataset.

Added an ML Insights tab with a Random Forest risk predictor (81.5% accuracy, 0.90 ROC-AUC), a LightGBM mortality forecaster (R^2 0.78 on held-out years), and an HDBSCAN country clustering model.

Book App & Reader Community Platform | *React, Backend APIs, Database Design* *In active development*

Full-stack platform where readers discover books and connect through a social community, with user interaction and content sharing. Frontend, backend APIs and database are being built together.

Technical Skills

Languages: Python, JavaScript, HTML/CSS, SQL
AI/ML: TensorFlow, Hugging Face Transformers, XGBoost, LightGBM, Random Forest, GPT-2 fine-tuning
Frameworks & Databases: React.js, Node.js, Flask, Django, FastAPI, MongoDB, PostgreSQL, Supabase
Tools: Render, Vercel, Git, Docker, Postman, Figma

Education

University of Ghana – B.S. Computer Engineering Accra, Ghana
Sep. 2021 – Nov. 2025